

Energy System Development in the Weight Room: Incorporating Prescribed Rest Periods for NCAA Men's Basketball Players

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ABSTRACT

TRAINING PROGRAMS OFTEN VARY SETS AND REPS TO ACHIEVE DIFFERENT TRAINING GOALS. REST IS NOT OFTEN PROGRAMMED WITHIN AN ANNUAL TRAINING CYCLE BUT CAN BE PRESCRIBED IN RESISTANCE TRAINING SESSIONS TO FURTHER DEVELOP ENERGY SYSTEMS DEMANDED BY SPORT. SPECIFICALLY, BASKETBALL INCLUDES INTERMITTENT PERIODS OF MODERATE TO HIGH INTENSITY, DERIVING MOST OF ITS ENERGY FROM THE ADENOSINE TRIPHOSPHATE-PHOSPHOCREATINE AND ANAEROBIC GLYCOLYTIC SYSTEMS. BY IDENTIFYING SPECIFIC WORK:REST RATIOS IN BASKETBALL, STRENGTH COACHES CAN PRESCRIBE LOGICAL REST PERIODS WITHIN A WEIGHT TRAINING SESSION. NOT

ONLY WILL THIS ENHANCE OVERALL ENERGY SYSTEM DEVELOPMENT, BUT WILL ALSO PROVIDE MORE STRUCTURE TO STRENGTH TRAINING ROUTINES.

INTRODUCTION

In the world of strength and conditioning, resistance training qualities are often manipulated to drive different adaptations and responses. Most typically, sets and reps are varied across programs in a periodized manner shifting toward lower reps during competition seasons and higher reps in the off-season. Rest prescription is not often addressed within a resistance training session. Most program writing may include sets, reps, exercise order, and intensity. Although there are limitations, programming intrasession rest can aid in training the energy system demanded by sport.

From both practical experience to physiological science, more rest is required between sets of low repetition and high

intensity. Creatine phosphate can deplete up to 50–70% during high-intensity exercise lasting between 5 and 30 seconds (5). In addition, it is well understood that a period of 3–5 minutes is required for the complete recovery of the adenosine triphosphate-phosphocreatine (ATP-PC) energy system (5,9). This system is critical in fueling high-intensity effort for extremely short periods. Furthermore, we also know that the presence of lactate, and the subsequent lowering of muscle pH, induces fatigue (12). This is particularly crucial when lifting at high intensities and low reps, across multiple sets and repeated efforts.

To incorporate rest, the energy system demand of sport must be understood. This article will specifically investigate these areas in terms of men's college basketball; however, research from various

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levels of basketball will be examined. The overarching goal of this article is to provide a logical framework from which strength coaches from any sport can prescribe rest periods practically. In addition, using sport-specific work-to-rest ratios in the weight room can further supplement on court or on field conditioning.

BASKETBALL ENERGY SYSTEM DEMANDS

Basketball demands intermittent high-intensity efforts (7). It is comprised of repeated sprints, explosive jumps, changes of direction, and copious accelerations and decelerations. A high-intensity run may occur every 20 seconds or so of gameplay and is just one of an average 600+ movements that an athlete performs in a game. It has also been calculated that a change in movement type may occur every 2–3 seconds of live action (10,11).

Because of the high-intensity nature of the sport, athletes rely heavily on anaerobic metabolism (7). Specifically, the availability of ATP produced by phosphocreatine hydrolysis and anaerobic glycogenolysis (9). During short sprints, explosive jumps, or other movements of extremely short duration, ATP is primarily synthesized from phosphocreatine sources (6,9). However, anaerobic glycolysis can support sustained efforts between 15 and 60 seconds, and gameplay can be sustained for these durations (9).

Aerobic metabolism also plays a role in basketball (6). Over the course of an entire game (including stoppages, timeouts, and halftime), approximately 60% of time is spent in low-intensity or recovery activities (11). This allows the aerobic metabolism to resynthesize ATP and phosphocreatine, aid in the metabolism and clearance of lactate, as well as remove inorganic phosphate intracellularly (6). Clearly, recovery through the aerobic system also highlights the need for rest due to the repletion of the ATP system and removal of lactate and other intracellular substrates (6,11). Although the anaerobic system should be trained for basketball, rest and recovery can also be manipulated to create the desired training effects.

In summary, basketball requires predominantly anaerobic metabolism to fuel repeated sprints and explosive efforts that occur on average once every 21 seconds (11). These activities can range between 0 and 60 seconds, with most gameplay continuously extending on the higher end of the spectrum. Biomechanical analysis proves that hundreds of different movements occur throughout the duration of a game, and most last only a few seconds at a time (10). Although training focuses primarily on the anaerobic energy system, prescribed recovery periods during training can enhance overall energy system development.

WORK:REST RATIOS IN BASKETBALL

Because of the anaerobic nature of the sport, a work:rest ratio of the ATP-PC system can be formulated. It is known that creatine phosphate depletes during high-intensity exercise lasting between 5 and 30 seconds, whereas complete ATP resynthesis occurs within 3–5 minutes (5). Thus, for an exercise bout lasting 30 seconds and a recovery of at least 180 seconds, a work:rest ratio for the ATP-PC system can be calculated as 1:6.

However, because of the intermittent style of gameplay, identifying one specific work:rest ratio in basketball is a difficult task. One can simply look at the number of minutes in a game compared with the nonlive minutes throughout a game event. To clarify, two 20-minute halves comprise a Division I men's basketball game. This equates to 40 minutes of live action if an athlete were to play the entire game.

On average, a collegiate basketball contest also includes a 15-minute halftime, 4 media timeouts per half, and, on average, 3 coaching timeouts ranging from 30 seconds to 1 minute. In addition, there is on average a 1:1 ratio between stoppage time and live action time, adding an additional 40 minutes (4). This equates to approximately 80 minutes of the game that is not live action. With a 40-minute game, and 80 minutes where the ball is not

bouncing, a work:rest ratio of 1:2 is justified for Division I men's basketball (Table 1).

When compared with research among different levels of basketball, this ratio seems to be relatively low. A more specific work:rest ratio of 1:3.6 was recorded among junior elite players (1). Furthermore, the intensity of the game and situation will also change the amount of rest required. Although some studies show that medium bouts of intensity drew a 1:1 ratio, multiple researchers have recorded ratios of 1:10 for extremely intense and maximal efforts during basketball competitions (1,3,8,11). When comparing light, to moderate, to heavy intensities, it was found that there was a ratio of 1:4:5 (2). What is clear is that the game is extremely variable. Using a broad scope, we can discern that the work:rest ratio is at least 1:2, but on average is probably higher.

A strength and conditioning coach should always adhere to the goals and specific demands of his or her team(s). Thus, it is advisable to open discussions with sport coaches regarding work:rest ratios, style of play desired, as well as overall team needs. Rest prescription can be individualized for teams and their style of play, and even further developed among individual players or specific positions. A strength and conditioning professional must balance research with practical application.

TIME UNDER TENSION

To apply specific rest periods, total time under tension (TUT) of a given working set must be understood. To do so, repetition tempo prescriptions can be used to calculate the total time of each set.

For example, an athlete is to perform each repetition of an exercise after a 4-0-1-0 tempo prescription. The tempo reads as follows: "4" is the eccentric portion of the rep, the first "0" is the time between eccentric and concentric portions, the "1" is the concentric contraction, and the second "0" is the time between reps. Using the above

Table 1
A comparison of nonlive time versus live gameplay in an NCAA Division I men's basketball game

Nonlive time		Gameplay	
3 × 30-s timeouts per half	3 min	First half:	20 min
1 × 60-s timeouts per half (avg.)	2 min	Second half:	20 min
4 × 165-s media timeouts per half (avg.)	22 min		
Halftime:	15 min		
Stoppage:live time	40 min		
41 min × 2 halves = 82 min		40 min	

example, each rep takes approximately 5 (4 + 0 + 1 + 0) seconds to perform. Then, to calculate the TUT, multiply the time of each rep by the number of reps.

For these purposes, the TUT of a set will be used synonymously with “work.” In other words, if TUT is 30 seconds, then 30 seconds of work is performed. Using this information and the ratios outlined in Table 2, work:rest ratios can be prescribed across sets of resistance training exercises.

PRESCRIBING REST

Rest should be prescribed in a logical and periodized fashion, given it is one of the training variables that can be manipulated within a program. When determining rest, factors such as exercise complexity, exercise intensity, as well as the overall training goal must be considered.

The more intense or more complex the exercise, the higher work:rest ratio

prescribed. That is to say, Olympic movement variations will most likely require the largest ratios, keeping in mind that the movements are powerful and TUT is short, while accessory exercises will require the smallest ratios. Furthermore, the closer a lift is to an athlete's 1 repetition maximum (RM), the more rest will be required. For sport, we are often working in the 1–10 rep range and between 75 and 100% 1RM (13). This would adhere to moderate-intensity:high-intensity zones, and it can be assumed that basketball work:rest ratios will fall somewhere between 1:2 and 1:10, through the discussion above.

Furthermore, bilateral exercises are inherently more complex and will require longer recovery periods than unilateral exercises (13). Also, keep in mind that during unilateral exercises, one side is resting while the other works, requiring less rest time from exercise to exercise. Barbell exercises are more neurologically complex than

dumbbell or machine exercises and thus will also require more rest.

Rest prescriptions can get even more specific for desired training goals. For example, if hypertrophy is the goal, work:rest ratios can be smaller. In other words, incomplete rest may be prescribed to induce more fatigue for a hypertrophy workout. Suggested rest periods for hypertrophy are between 30 and 90 seconds (13). On the other hand, if the desired training goal is relative strength or power, the higher intensities per set would induce longer rest periods and therefore larger work:rest ratios. The total amount of rest between sets would fall somewhere between 2 and 5 minutes (13).

Finally, rest can be manipulated much like volume and intensity within an annual training cycle. Depending on the time of year, different strength qualities are trained. Collegiate basketball can be broken into 4 periodization periods: off-season, preseason, in-season, and postseason. Following a linear model, work:rest ratios can be altered to further enhance training goals. Table 3 illustrates a sample work:rest progression over an annual training cycle.

APPLICATION OF REST

To make rest ratios more applicable, resistance training sessions are broken down into primary, assistance, and accessory blocks. The primary block will most closely adhere to specific work:rest ratios, whereas assistance and accessory exercises no longer require prolonged rest periods. For basketball, most intralift rest prescriptions outside of the season will look like the following:

- A Block (Primary): 90- to 120-second rest.
- B Block (Assistance): 60- to 90-second rest.
- C Block (Accessory): 30- to 45-second rest.

To help illustrate these points, an example lower-body strength training session and an upper-body hypertrophy session are provided in Figures 1 and 2.

Table 2
A summary of work:rest ratios from author and research

Source	Work:rest ratio
Division I NCAA men's basketball game	1:2
Ben Abdelkrim et al. (1)	1:3.6
Bishop and Wright (2)	1:4-5
ATP-PC energy system	1:6
Ben Abdelkrim et al. (1), Buchheit (3), and McInnes et al. (11)	1:10

Table 3
Example collegiate basketball linear macrocycle with work:rest prescription

Annual training macrocycle				
Period:	Postseason	Off-season	Preseason	In-season
Time of year	March–May	May–August	August–October	October–March
Phase:	Hypertrophy/endurance	Hypertrophy/basic strength	Basic strength/power	Strength/power/peak
Intensity:	60–75% 1RM	75–95% 1RM	80–95% 1RM	75–100% 1RM ^a
Work:rest	1:2	1:2–1:3	1:3–1:5	1:6–1:10
^a Broad range to account for high volume of sport.				
RM = repetition maximum.				

Often, it is impractical to time each athlete's specific rest period. Many collegiate programs have multiple athletes training together, and it is important to note that the NSCA recommended coach to athlete ratio for collegiate athletics is 1:20 (14). To circumvent this situation, athletes can perform sets in groups. Essentially, the team would be broken into as many groups as the desired work:rest ratio. If the desired ratio is 1:3, split the team into 4 groups. One group performs the lift, while the other 3 rest. For a superset, groups would lift one at a time before moving onto the next exercise.

For assistance and remedial exercises (B and C blocks), rest is less important. That being said, some B blocks require more rest. For instance, if primary, multijoint exercises are still performed

in B series, it is recommended that athletes rest and allow for the energy systems to recover before performing their next set.

It is important to make the addition of rest periods realistic. The focus should be on the primary exercises and energy system recovery to allow for optimal strength and power output in a session, as well as the development of specific energy systems. Rest can also be applied to assistance and accessory exercises as needed, but this is a portion of the training session where time can be condensed, and athletes can move between exercises more quickly. Instead of athletes completing sets and reps at their convenience, more structure and applicable science can be injected into a training session by designating rest periods.

PRACTICAL LIMITATIONS TO REST PRESCRIPTION

Time and space. Understandably, not all situations allow for athletes to have optimal time allowing intrasession rest periods. This may be due to weight room traffic, limited equipment and space, or restrained time through sport coaches. This may also be the case for coaches with a high number of athletes training at one time. If this is the case, focus on implementing rest periods for primary exercises only, while being conscious of set and rep prescriptions.

Athlete and coach comprehension. For most athletes, especially basketball players, it is hard to understand taking time to rest. In their eyes, the game is all about sprinting and getting up and down the floor. This can also be a barrier for sport coaches, especially when walking into the weight room and seeing athletes not doing anything. Educating athletes and coaches is important. In terms of basketball, describing the game as intense efforts (running up and down the floor continuously) followed by rest due to a whistle or a timeout is justified.

Sport demands. Rest prescription in the weight room is not realistic for all sports and athletes. For example, baseball or softball athletes have very large work:rest ratios, and thus, it would become impractical to mimic these in a resistance training session. In addition, individual sports, such as swimming and

Lower-Body Strength Routine				
A Block	Exercise	Sets x Reps	Tempo	Rest
A1	Barbell Back Squat	5x5	4-0-1-0	120 s
A2	Prone Leg Curl (Bilateral)	5x5	4-0-1-0	120 s
B Block	Exercise	Sets x Reps	Tempo	Rest
B1	Front Foot Elevated DB Split Squat	4x6 ea.	3-0-1-0	60 s*
B2	Barbell Romanian Deadlift	4x6	3-0-1-0	60 s*
C Block	Exercise	Sets x Reps	Tempo	Rest
C1	Lateral DB Step-up	3x8	2-0-1-0	30 s**
C2	Standing Calf Raise	3x12	2-0-1-0	30 s

Figure 1. Example lower-body strength routine, including rest intervals. *Rest is 60 seconds because B1 is a unilateral exercise. **Rest is 30 seconds because C1 is a unilateral exercise.

Upper-Body Hypertrophy Routine				
A Block	Exercise	Sets x Reps	Tempo	Rest*
A1	Barbell Overhead Press	5x8	3-0-1-0	90 s
A2	Chin-up (Supinated Grip)	5x8	3-0-1-0	90 s
B Block	Exercise	Sets x Reps	Tempo	Rest*
B1	Flat DB Bench Press	4x10	3-0-1-0	60 s
B2	Seated Cable Row	4x10	3-0-1-0	60 s
C Block	Exercise	Sets x Reps	Tempo	Rest*
C1	Seated, Neutral Grip DB Curl	3x15	2-0-1-0	30 s
C2	EZ Bar French Press	3x15	2-0-1-0	30 s

Figure 2. Example upper-body hypertrophy routine including designated rest periods. *Rest is on the low end of the range for each block because it is a dedicated hypertrophy session.

track and field are comprised of various events and different energy requirements.

CONCLUSION

Basketball is an intermittent game comprising various movements and short-duration sprints (7). The primary energy systems used are the ATP-PC and anaerobic glycolytic systems (6,7,9). Aerobic metabolism may also play a small role as studies have shown a large part of the game is nonlive action (6,11). Because of this, it is hard to justify a specific work:rest ratio for basketball. With various research across all levels of basketball, and a comparison of live action and non-live action gameplay in a men's NCAA Division I game, work:rest ratios for the sport have been identified somewhere between 1:2 and 1:10 (1–3,11).

To quantify work, muscular TUT is accounted for within a resistance training session. Exercise nature and complexity, as well as time of year in the annual training program, will further give insight into a practical way to apply work:rest ratios. In addition, specific training goals will require different amounts of rest and alter these ratios.

Ultimately, rest is just one of many various training principles that can be altered per athlete or specific training goal. Furthermore, by understanding the energy requirements of sport, one can further enhance energy system

training in the weight room, getting outside of traditional conditioning and agility sessions. Rest can be varied per exercise, training goal, or specific energy system development. Some limitations to applying rest periods within a training session are time and space, athlete and coach comprehension, as well as various sport and athlete demands.

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REFERENCES

1. Ben Abdelkrim N, Castagna C, Jabri I, Battikh T, El Fazaa S, and El Ati J. Activity profile and physiological requirements of junior elite basketball players in relation to aerobic-anaerobic fitness. *J Strength Cond Res* 24: 2330–2342, 2010.
2. Bishop DC and Wright C. A time-motion analysis of professional basketball to

determine the relationship between three activity profiles: High, medium and low intensity and the length of the time spent on court. *Int J Perform Anal Sport* 6: 130–139, 2006.

3. Buchheit M. The 30-15 intermittent fitness test: Accuracy for individualizing interval training of young intermittent sport players. *J Strength Cond Res* 22: 365–374, 2008.
4. Conte D, Tessitore A, Smiley K, Thomas C, and Favero TG. Performance profile of NCAA Division I men's basketball games and training sessions. *Biol Sport* 33: 189, 2016.
5. Carmer J, Herda T, Haff G, and Triplett NT. *Essentials of Strength Training and Conditioning*. Champaign, IL: Human Kinetics, 2015. pp. 43–63.
6. Glaister M. Multiple sprint work. *Sports Med* 35: 757–777, 2005.
7. Hoffman JR, Epstein S, Einbinder M, and Weinstein Y. The influence of aerobic capacity on anaerobic performance and recovery indices in basketball players. *J Strength Cond Res* 13: 407–411, 1999.
8. Hoffman JR and Maresch CM. Physiology of basketball. In: *Exercise and Sport Science*. Garrent WE and Kirkendall DT, eds. Philadelphia, PA: Lippicott Williams & Wilkins, 2000. pp. 733–744.
9. MacDougall D, MacDougall JD, and Sale D. *The Physiology of Training for High Performance*, Oxford, United Kingdom: Oxford University Press, 2014.
10. Matthew D and Delextat A. Heart rate, blood lactate concentration, and time-motion analysis of female basketball players during competition. *J Sports Sci* 27: 813–821, 2009.
11. McInnes SE, Carlson JS, Jones CJ, and McKenna MJ. The physiological load imposed on basketball players during competition. *J Sports Sci* 13: 387–397, 1995.
12. Metzger JM and Fitts RH. Role of intracellular pH in muscle fatigue. *J Appl Physiol* 62: 1392–1397, 1987.
13. Sheppard J, Haff G, and Triplett NT. *Essentials of Strength Training and Conditioning*. Champaign, IL: Human Kinetics, 2015b. pp. 439–469.
14. Statler T, Brown V, Haff G, and Triplett NT. *Essentials of Strength Training and Conditioning*. Champaign, IL: Human Kinetics, 2015. pp. 641–657.